BAYESIAN VALIDATION OF ATTRACTION BASINS FOR BLOCK MAXIMA EXTREME VALUE MODELS

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Selecting between the three bassins of attraction defining the class of block maxima models can be preferable than using the encompassing GEV family, to get more reliable estimates of extreme quantiles. In an informative Bayesian setting, prior elicitation has to be balanced with respect to the differences of complexity between the GEV submodels, while accounting for a similar (model-agnostic) interpretation of available prior information. This article defines a closed-form prior elicitation based on approximating posterior distributions build on virtual data and reference prior measures. This elicitation is shown to be appropriate for calibrating the priors in a compatible way, limiting the number of subjective parameters and the complexity bias threatening model selection through Bayes factors. Simulated and real situations are studied and highlighted in the corresponding talk.

Keywords: Extreme value models; Bayesian statistics; model selection; prior elicitation; virtual data

1. Introduction

Extreme value models applied to reliability analysis appear in various applications. A natural framework is the design and robustness study of industrial installations against natural hazards [1]. In a structural reliability framework where it is typically wished to compute a failure probability $P(g(X) \leq 0)$ [2] , with g a so-called computer model and X a random input vector of loads and forcings [3], extreme value models are also appropriate models for one or several dimensions X_i of X, in a context of conservative risk assessment (e.g., X_i may be the size of larger defects within a material). Other usual frameworks are for instance related to financial risk analysis or cybersecurity.

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In this article, we consider a Bayesian Block Maxima (BM) framework [4], that traduces the behavior of an univariate sample of maxima (or minima) $X_1, \ldots, X_n$ of a random phenomenon observed over a given period. In such a case, X asymptotically follows a Generalized Extreme Value (GEV) distribution, defined as an embedding model for the Fréchet, Gumbel and Weibull distributions. It is an alternative to the Peak-Over-Threshold (POT) framework. The sign and value of the GEV shape parameter $\xi \in \mathbb{R}$ are known to strongly influence the tails of this distribution and, beyond, the assessment of return levels and periods [1]. Denoting $\theta = (\mu, \sigma, \xi)$ the usual parameterization of the embedded distributions, for $\xi > 0$ the attraction domain of extreme values is Fréchet, corrresponding to probability distribution function

$$P(X < x|\theta) = \exp\left\{-\left(\frac{x-\mu}{\sigma}\right)^{-1/\xi}\right\}, \tag{1}$$

with $\sigma > 0$, $\mu \in \mathbb{R}$ and $x \geq \mu$, characterized by wide return levels and no upper bound. For $\xi = 0$, it is Gumbel, such that

$$P(X < x|\theta) = \exp\left\{-\exp\left(-\frac{x-\mu}{\sigma}\right)\right\} \tag{2}$$

with $\sigma > 0$, $\mu \in \mathbb{R}$ and $x \in \mathbb{R}$. For $\xi < 0$, the attraction domain is Weibull, being upperly bounded, such that

$$P(X < x|\theta) = \exp\left\{-\left(\frac{\mu-x}{\sigma}\right)^{1/\xi}\right\} \tag{3}$$

and with $\sigma > 0$, $\mu \in \mathbb{R}$ and $x \leq \mu$. As pointed in [5,6], finding the nature of the correct basin of attraction, then providing relevant credible intervals on return levels and periods, appears to be more related to selection between these three distribution rather than computing a direct estimator of ξ, accompanied with a confidence domains for ξ that cover both negative and positive values.

Following the advice of many authors, a Bayesian framework should be chosen to take into account the strong uncertainties characterizing extreme value problems, the possible availability of subjective prior information and the predictive nature of the prognosis provided by the use of such models [1] (Chapter 9). Hence a prior $\pi(\theta)$ should be elicited. Since it plays a key role in inference and model selection (e.g. using Bayes factors), the methodology of prior elicitation must be carefully chosen, based on formal rules or properties. While such a clear methodology was proposed by [7]

for the POT framework, there is still a lack for the BM framework. This article proposes to fill this space left vacant.

2. Modelling

2.1. *Elicitation principle*

Bayesian approaches still suffer from the usual criticism concerning the subjectivity of prior modeling, while the theoretical and practical benefits of this framework have been demonstrated for a long time [10]. It is besides well known that Bayes factors can strongly depend on prior effects. For these reasons, Bayesian statisticians have always to reinforce the methodologies of prior elicitation. The approach proposed here is based on exhibiting parametric prior structures that have meaningful (hyper)parameters and can be interpreted as approximate *posterior priors* conditional to *virtual data* and so-called noninformative priors. Roughly speaking, the prior hyperparameters are statistics of this virtual dataset, which can be calibrated from additional information. Recommended in [9], this approach allows to modulate prior information using the size of virtual data, associated to the strength of expert knowledge. It reduces to conjugate priors when the sampling model belongs to the natural exponential family (as the Gumbel model). Besides, this strategy offers the possibility, for the non-conjugate extreme models, to provide priors that offer a semi-conjugate structure.

Such a principle defines a prior form, provided for each model in next section. The calibration itself depends on the statistical interpretation of prior information. In accordance with the approach recommended by [11] (and by many other researchers [12]), prior guesses are provided with respect to the marginal distribution $f(x) = \mathrm{E}_\pi[f(x|\theta)]$ rather than with respect to the probability density function $f(x|\theta)$ of each model in competition; there are therefore interpreted as estimators of prior predictive quantiles.

Finally, the sense given to prior hyperparameters allows to build a rule of prior compatibility, in view of avoiding modeling bias in model selection.

2.2. *Main technical results*

We extend here the result obtained by [14] in the special case of the Gumbel distribution. A conjugate prior for Gumbel distribution is the following:

$$\pi(\mu,\sigma) \propto \sigma^{-m} \exp\left(m\frac{(\mu-\bar{\tilde{x}}_m)}{\sigma} - \sum_{i=1}^{m} \exp\left\{-\frac{\tilde{x}_i-\mu}{\sigma}\right\}\right) \tag{4}$$

which is proper provided $m \geq 3$ (Theorem 1 in [14]), and where hyperparameters $(m, \bar{\tilde{x}}_m, \tilde{x}_1, \ldots, \tilde{x}_m)$ correspond respectively to the size of a prior virtual sample, its mean and the virtual data themselves. Given true observations $\mathbf{x_n}$, the posterior distribution can be written as

$$\pi(\mu,\sigma|\mathbf{x_n}) \propto \sigma^{-m-n} \exp\left(\{m+n\}\frac{\left(\mu - \frac{m\bar{\tilde{x}}_m + n\bar{\mathbf{x}}_\mathbf{n}}{m+n}\right)}{\sigma} \right.$$
$$\left. - \sum_{i=1}^{m} \exp\left\{-\frac{\tilde{x}_i-\mu}{\sigma}\right\} - \sum_{k=1}^{n} \exp\left\{-\frac{x_k-\mu}{\sigma}\right\}\right).$$

Roughly speaking, these extensions are summarized in the next results.

Reparametrize the Fréchet distribution:

$$P(X<x|\theta) = \exp\left\{-\nu\,(x-\mu)^{-1/\xi}\right\}$$

and denote now $\theta = (\mu, \nu, \xi)$ with $\nu = \sigma^{1/\xi} > 0$. Denote $\mathbb{G}(a,b)$ the gamma distribution with mean a/b and shape parameter a, and denote $\mathbb{IG}(a,b)$ the corresponding inverse gamma distribution.

Proposition 2.1. *Assume the Fréchet prior distribution $\pi(\nu,\mu,\xi)$ defined by*

$$\begin{aligned} \nu|\mu,\xi &\sim \mathbb{G}\left(m, s_1(\mu,\xi)\right), \\ \xi|\mu &\sim \mathbb{IG}\left(m, s_2(\mu)\right), \\ \pi(\mu) &\propto \frac{\mathbb{1}_{\{\mu\leq x_{e_1}\}}}{(x_{e_2}-\mu)^m s_2^m(\mu)} \end{aligned} \tag{5}$$

where $\mu < x_{e_1} < x_{e_2}$ and

$$\begin{aligned} s_1(\mu,\xi) &= m(x_{e_1}-\mu)^{-1/\xi}, \\ s_2(\mu) &= m\log\left(\frac{x_{e_2}-\mu}{x_{e_1}-\mu}\right). \end{aligned}$$

Then $\pi(\nu,\mu,\xi)$ is conjugated for ν given (μ,ξ), and when $m \in \mathbb{N}^*$, $\pi(\nu,\mu,\xi) = \pi^R(\nu,\mu,\xi|\tilde{\mathbf{x}}_\mathbf{m})$ where π^R is the Fréchet reference prior and

$\tilde{\mathbf{x}}_{\mathbf{m}}$ *is a virtual Fréchet sample of size* m *with statistics* $\{x_{e_2}, x_{e_3}\}$*, interpretable as the (shifted) geometric and inverse arithmetic means of these virtual data.*

A semi-conjugate prior for Weibull can be partly established by obvious symmetry with the Fréchet case, adapting the ideas expressed in [8] and [9]. Using the same parametrization,

$$P(X < x|\theta) = \exp\left\{-\nu\left(\mu - x\right)^{1/\xi}\right\}.$$

Proposition 2.2. *Assume the Weibull prior distribution* $\pi(\nu, \mu, \xi)$ *defined by*

$$\begin{aligned} \nu|\mu, \xi &\sim \mathbb{G}\left(m, s_3(\mu, \xi)\right), \\ \xi|\mu &\sim \mathbb{IG}\left(m, s_4(\mu)\right), \\ \pi(\mu) &\propto \frac{\mathbb{1}_{\{\mu \geq x_{e_4}\}}}{(\mu - x_{e_4})^m s_4^m(\mu)} \end{aligned} \tag{6}$$

where $x_{e_4} > x_{e_3}$ *and*

$$\begin{aligned} s_3(\mu, \xi) &= m(\mu - x_{e_3})^{-1/\xi}, \\ s_4(\mu) &= m \log\left(\frac{\mu - x_{e_3}}{\mu - x_{e_4}}\right). \end{aligned}$$

Then $\pi(\nu, \mu, \xi)$ *is conjugated for* ν *given* (μ, ξ)*, and when* $m \in \mathbb{N}^*$*,* $\pi(\nu, \mu, \xi) = \pi^R(\nu, \mu, \xi|\tilde{\mathbf{x}}_{\mathbf{m}})$ *where* π^R *is the Weibull reference prior and* $\tilde{\mathbf{x}}_{\mathbf{m}}$ *is a virtual Weibull sample of size* m *with similar statistics* $\{x_{e_3}, x_{e_4}\}$*.*

2.3. *Marginal calibration*

For each model, given a virtual size m, the calibration of virtual summary stastistics can be conducted using various loss functions, as Cooke's criterion. An instance of such a calibration result is provided in Table 2.3. This table provides hyperparameters for the Fréchet prior (under the constraint $\mu \geq 0$) and computation of the effective orders of prior predictive quartiles, which must be compared to ordered specifications of $(25\%, 50\%, 75\%)$

2.4. *Compatibility principle for complete calibration before model selection*

Although intuitive at first sight, choosing a similar virtual size m for each model appears not suitable to address a fair Bayesian model comparison, in the sense defined by [15,13]. Indeed, consider models i and j with smaller

Virtual size m	x_{e_1}	x_{e_2}	Order of prior predictive quartiles (75,100,150)
1	100.41	130.20	$[25\%, 50\%, 75\%]$
2	95.30	138.39	$[24\%, 49\%, 74\%]$
3	91.22	136.93	$[23\%, 51\%, 74\%]$
4	89.18	135.10	$[24\%, 50\%, 74\%]$
5	87.72	133.95	$[24\%, 51\%, 75\%]$
6	87.65	133.88	$[24\%, 50\%, 75\%]$
7	87.14	133.26	$[25\%, 50\%, 74\%]$
10	86.63	132.65	$[25\%, 51\%, 75\%]$
15	85.11	132.24	$[26\%, 50\%, 75\%]$

and larger complexities (approximately driven by the number of free parameters). Since model i is simplest to infer on from the same amount of information, the corresponding virtual size required to transport this amount of information is smaller than the virtual size required by model j to transport the same amount of information. Reusing the notion of prior compatibility chosen by [16], it is possible to define a strategy for calibrating the virtual sizes by an equivalence rule between models. This strategy stands on the fact that a parametric model f_a with prior π_a is nested in another parametric model f_b, accompanied by prior π_b. It is expected that the Kullback-Leibler information-loss between the more flexible marginal model g_b and the constrained marginal model g_a:

$$KL\left(g_b|g_a\right) = \int g_b(x) \log \frac{g_b(x)}{g_a(x)} \, dx$$

where

$$g_c(x) = \int_\Theta g_c(x|\theta)\pi_c(\theta) \, d\theta,$$

be minimized when the priors are compatible. This minimization can be conducted only on virtual sizes only, limiting the impact of subjective procedures. Results and perspectives will be presented during the talk.

3. Conclusion

This article summarizes research results allowing, for the Bayesian practitioner of extreme value models, to define justifiable, coherent prior forms, by assimilating prior information to virtual data information (at least approximately). Such a strategy is an extension of the usual conjugate strategy, which cannot be directly applied in the Block Maxima framework.

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